MIL-E-0022200/2B(SHIPS)
3 January 1969
USED IN LIEU OF
MIL-E-22200/2A
29 June 1962
(See 6.5)

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED (AUSTENITIC CHROMIUM-NICKEL STEEL, FOR CORROSIVE AND HIGH TEMPERATURE SERVICES)

This limited coordination Military specification has been prepared by the Department of the Navy, Naval Ship Engineering Center based upon currently available technical information, but it has not been approved for promulgation as a revision of Military Specification MIL-E-22200/2A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers shielded metal arc welding covered electrodes, for manually depositing austenitic chromium-nickel steel weld metal.

1.2 Classification. The electrodes shall be of the following types, classes and sizes, shown in table I as specified (see 6.2.1):

Table I - Classification

Type	Class	Positions of welding	Sizes (diameter) inch
MIL-308-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-308-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-308L-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-308L-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-308HC-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-308HC-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-309-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-309-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-309L-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-309L-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-309Cb-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-309Cb-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-310-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-310-16	3	Flat	5/16
MIL-312-15	1	All	5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-312-16	3	Flat	5/16

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Table I - Classification (cont'd)

Type	Class	Positions of welding	Sizes (diameter) inch
MIL-316-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-316-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-316L-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-316L-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-317-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-317-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-318-15	1	All	3/32, 1/8, 5/32
MIL-318-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-330-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-330-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-347-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-347-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-347HC-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-347HC-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-349-15	1	All	1/16/ 5/64, 3/32, 1/8, 5/32
MIL-349-16	2	Horizontal fillets and flat	3/16, 1/4
MIL-16.8.2-15	1	All	1/16, 5/64, 3/32, 1/8, 5/32
MIL-16.8.2-16	2	Horizontal fillets and flat	3/16, 1/4

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-766 - Steel Plates, Sheets, and Strip-Corrosion Resisting.

MILITARY

MIL-W-10430 - Welding Rods and Electrodes: Preparation for Delivery of
MIL-E-0022200E - Electrodes, Welding, Covered, General Specification for

STANDARDS

MILITARY

MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base

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STANDARDS (cont'd)

MILITARY
MIL-STD-271 - Nondestructive Testing Requirements for Metals
MIL-STD-418 - Mechanical Tests for Welded Joints

PUBLICATIONS

MILITARY
NAVSHIPS 0900-003-9000 - Radiographic Standards for Production and Repair Welds

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A167 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet and Strip
A240 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Plate, Sheet and Strip for Fusion-Welded Unfired Pressure Vessels

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103).

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Electrodes furnished under this specification shall conform to the requirements of MIL-E-0022200E and as specified hereinafter.

3.2 Coverings.

3.2.1 The chemical composition of the electrode covering is optional with the manufacturer subject to the requirements of 3.4 of MIL-E-0022200E. The total water content in the covering of each electrode type shall be determined as a qualifying test, and moisture result obtained thereby shall be used to establish the maximum moisture level for this electrode under this specification. The maximum moisture content established for each electrode type qualified shall be reported in the certification of quality conformance tests, figure 5.

3.3 Groove Welds.

3.3.1 Mechanical properties. The mechanical properties of groove welds shall be as specified in table II.

3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table III.

3.5 Electrode identification. Electrodes shall be identified as specified in 3.5.10 of MIL-E-0022200E.

3.6 Nominal lengths. The specified nominal lengths of electrodes shall be in accordance with table IV.

3.7 Decoated core wire. Any core wire or electrode which is removed from the assembly line area during production of a lot of electrodes shall not be used to fill any order under this specification.

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Table II - Mechanical properties of deposited weld metal

Type	Tensile strength (min.)	Elongation[1/] in 2 inches (min.)	Bend		Fissures[2/3/4/] per bend specimen tested (max.)
			As-welded	Heat-treated	
	P.s.i.	Percent			
MIL-308-15 and 16	80,000	35	X	—	3
MIL-308L-15 and 16	75,000	35	X	—	2
MIL-308HC-15 and 16	82,000	35	X	—	3
MIL-309-15 and 16	80,000	35	X	—	2
MIL-309L-15 and 16	75,000	35	X	—	2
MIL-309Cb-15 and 16	80,000	30	X	—	2
MIL-310-15 and 16	80,000	30	X	—	4
MIL-312-15 and 16	95,000	22	X	—	1
MIL-316-15 and 16	80,000	30	X	—	3
MIL-316L-15 and 16	75,000	30	X	—	3
MIL-317-15 and 16	80,000	30	X	—	3
MIL-318-15 and 16	80,000	25	X	—	2
MIL-330-15 and 16	75,000	30	X	—	4
MIL-347-15 and 16	80,000 as-welded, and as heat-treated	30 as-welded, and 25 as heat-treated	X	X	2
MIL-347HC-15 and 16	82,000 as-welded and as heat-treated	30 as-welded and 25 as heat-treated	X	X	3
MIL-349-15 and 16	100,000	25	X	—	3
MIL-16.8.2-15 and 16	80,000 as-welded and as heat-treated	35 as-welded, and as heat-treated	X	X	2

[1/] Elongation not required for transverse weld specimens.

[2/] Bend test specimens or coupons shall be visually examined on the convex face by the unaided eye (corrected if necessary to 20/20 vision) and shall not exhibit more than the specified number of fissures after completion of bending. The presence of a single tear or fissure having a dimension larger than 1/16-inch measured in any direction shall be cause for rejection or failure under test.

[3/] Fissures 1/64-inch and less shall not be counted.

[4/] The corners of specimens shall not be considered when evaluating the results.

4. QUALITY ASSURANCE PROVISIONS

4.1 Comparison inspection. In addition to the examination and tests required in 4.5 of MIL-E-0022200E, tests specified in table V herein shall be conducted as specified in 4.3 of MIL-E-0022200E, when required by the Naval Ship Engineering Center (Metals Section).

4.2 Quality Conformance Inspection.

4.2.1 Lot. A lot of electrodes shall be as specified in 4.4.2 and 4.4.6 of MIL-E-0022200E for inspection level A.

4.2.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with 4.4.7 through 4.4.8.3 of MIL-E-0022200E.

4.2.3 Quality conformance inspection. The samples selected in accordance with 4.2.2 herein shall be subjected to the tests specified in Table VI herein.

4.2.3.1 Unsatisfactory test results. If the results of the first test are determined to be unsatisfactory, two retests shall be permitted. The results of both retests shall be satisfactory for lot acceptance.

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Table III - Chemical composition (percent) of deposited weld metal

Type	Percentage (single values are maximum)									
	Carbon	Manganese	Phosphorus	Sulfur	Silicon	Chromium	Nickel	Molyb-denum	Colum-bium	Tung-sten
1/ MIL-308-15 and 16	0.08	2.5	0.04	0.03	0.90	18.0-21.0	9.0-11.0	---	---	---
1/ MIL-308L-15 and 16	.04	2.5	.04	.03	.90	18.0-21.0	9.0-11.0	---	---	---
2/ MIL-308HC-15 and 16	.08-.15	2.5	.03	.03	3/	18.0-21.0	9.0-11.0	---	---	---
1/ MIL-309-15 and 16	.10	2.5	.04	.03	.90	22.0-25.0	12.0-14.0	---	---	---
1/ MIL-309L-15 and 16	.04	2.5	.04	.03	.90	22.0-25.0	12.0-14.0	---	---	---
1/ MIL-309Cb-15 and 16	.12	2.5	.04	.03	.90	22.0-25.0	12.0-14.0	---	Cb&Ta .70-1.00	---
MIL-310-15 15 and 16	.20	2.5	.03	.03	.75	25.0-28.0	20.0-22.0	---	---	---
MIL-312-15 and 16	.15	2.5	.04	.03	.90	28.0-32.0	8.0-10.5	---	---	---
MIL-316-15 and 16	.08	2.5	.04	.03	.90	17.0-20.0	11.0-14.0	2.0-2.5	---	---
MIL-316L-15 and 16	.04	2.5	.04	.03	.90	17.0-20.0	11.0-14.0	2.0-2.5	---	---
MIL-317-15 and 16	.08	2.5	.04	.03	.90	18.0-21.0	12.0-14.0	3.0-4.0	---	---
MIL-318-15 and 16	.08	2.5	.04	.03	.90	17.0-20.0	11.0-14.0	2.0-2.5	4/	---
MIL-330-15 and 16	.25	2.5	.04	.03	.90	14.0-17.0	33.0(min.)	---	---	---
1/ MIL-347-15 and 16	.08	2.5	.04	.03	.90	18.0-21.0	9.0-11.0	---	5/	---
MIL-347HC-15 and 16	.08-.15	2.5	.025	.025	3/	18.0-21.0	9.0-11.0	---	6/	---
MIL-349-15 and 16	.07-.13	2.5	.03	.03	1.00	18.5-21.0	8.0-10.0	.35-.65	7/0.75-1.2	1.25-1.75
MIL-16.8.2-15 and 16	.10	2.5	.025	.025	.50	14.5-16.5	7.5-9.5	1.0-2.0	---	---

1/ Delta ferrite content shall be from 4 to 9 percent (determined by Magne-Gage). (See figure 4 and notes).

2/ Delta ferrite content shall not exceed 4 percent (determined by Magne-Gage). (See figure 4 and notes).

3/ Silicon shall be not more than 5 times actual carbon present.

4/ Columbium (combined CB+Ta) shall be not less than 6 times actual carbon present but not more than 0.80 percent

5/ Columbium (combined Cb+Ta) shall be not less than 10 times actual carbon present but not more than 1.00 percent. Tantalum shall not exceed 0.10 percent.

6/ Columbium (combined Cb+Ta) shall be not less than 8 times actual carbon present but not more than 1.4 percent

7/ Titanium shall be limited to 0.15 percent maximum.

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Table IV - Nominal lengths.

Sizes	Electrode lengths end grips
Inch	Inches
1/16, 5/64	9
3/32	9 or 12
1/8	9 or 14
5/32 to 1/4 inclusive	14
5/16	18

Table V - Comparison inspection tests

Electrode types	Sizes	Test	Requirements
	Inch		
All	All	4.6.8 (MIL-E-0022200E)	3.4, table III herein
All	3/32 through 5/16	Weld test No. 3 (see table VII herein)	3.7.1 (MIL-E-0022200E) and tables II and III herein

Table VI - Summary of tests required for quality conformance inspection.

Type	Required Tests			Procedures & Requirements
	Chemical Analysis[1/]	Ferrite Content Test No. 4	Groove Weld Usability Test No. 8a	
MIL-308	Cr, Ni	X	X	For Chemistry use 4.6.8 of MIL-E-0022200E
MIL-308L	C, Cr, Ni	X	X	
MIL-308HC	C, Cr, Ni	X	X	
MIL-309	Cr, Ni	—	X	
MIL-309L	C, Cr, Ni	—	X	
MIL-309Cb	Cr, Ni, Cb	—	X	For ferrite content, see Table VII and notes thereto
MIL-310	Cr, Ni	—	X	
MIL-312	Cr, Ni	—	X	
MIL-316	Cr, Ni, Mo	—	X	
MIL-316L	C, Cr, Ni, Mo	—	X	
MIL-317	Cr, Ni, Mo	—	X	
MIL-318	C, Cr, Ni, Mo, Cb	—	X	
MIL-330	Cr, Ni	—	X	
MIL-347	Cr, Ni, Cb	X	X	
MIL-347HC	C, Cr, Ni, Cb	—	X	
MIL-349	Cr, Ni, Mo, Cb, W	—	X	For Weld Test 8a, See Table VII
MIL-16.8.2	Cr, Ni, Mo	—	X	
All	Diameter and length of core wire Concentricity			Table VIII

[1/] Required only for elements indicated in accordance with limits specified by Table III.

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4.3 Test Procedures

4.3.1 Weld Tests. Table VII summarizes the weld tests required for qualification testing of electrodes under this specification (see 4.2 of MIL-E-0022200E) and for inspection testing as specified in Tables V and VI herein.

Table VII - Summary of required weld tests[1/]

Weld test No.	Electrode Type	Diameter Sizes 1/16	5/64	3/32	1/8	5/32	3/16	1/4	5/16	Weld Joint Design Procedure	Welding Position[2/]	Tests: X-ray	Tensile	Bend	Chemical analysis[3/]	Ferrite content[4/]	Requirements
3	All Types	-	-	X	X	X	X	X	X	Figure 1 and Notes	F	X	X	X	X	—	3.7.1 of MIL-E-0022200E and Tables II and III herein. 4.6.8 of MIL-E-0022200E
6	All Types	X	X	-	-	-	-	-	-	Figure 2 and Notes	F	X	X	-	X		3.7.1 of MIL-E-0022200E and Tables II and III herein. 4.6.8 of MIL-E-0022200E
8a	All Types	X	X	X	X	X	-	-	-	Figure 3 and Notes	V	X	-	-	-	-	3.7.1 of MIL-E-0022200E
		-	-	-	-	-	X	X	X		F	X[5/]	-	-	-	-	
4	MIL-308, 308L, 308HC 347,309, 309L,309Cb	X	X	X	X	X	X	X	X	Figure 4 and Notes	F	-	-	-	-	X	Table III herein

[1/] For qualification testing of the electrodes for the Qualified Products List (QPL), all electrode types shall be tested using direct current reverse polarity. In addition, MIL-XXX-16 electrode types shall be tested using alternating current. For comparison inspection and quality conformance inspection, all electrode types shall be tested using direct current reverse polarity.

[2/] F - flat V - vertical

[3/] Chemical analysis shall be obtained by the procedures and methods described in 4.6.8 of MIL-E-0022200E, except the ferrite test pad may be used in lieu of the weld pad required in 4.6.8 (MIL-E-0022200E) for obtaining chips for those electrode types requiring ferrite determination.

[4/] Ferrite content of welds shall be determined by Magne-gage (Aminco-Brenner) or equivalent, calibrated against specimens having known ferrite contents.

[5/] For QPL and comparison inspection testing, the 8a test may be eliminated for 3/16, 1/4, and 5/16-inch diameter electrodes.

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4.3.2 Other tests. Table VIII summarizes tests, in addition to those specified in table VII, required for qualification testing of electrodes for the Qualified Products List (QPL), comparison inspection and quality conformance inspection, as applicable.

Table VIII - Summary of other tests.

Tests	Procedures	Requirements
	MIL-E-0022200E	
Diameter and length of core wire	4.6.2.3	3.10 (MIL-E-0022200E) and 3.6 herein
Concentricity	4.6.2	3.5.2 (MIL-E-0022200E)
Dielectric strength[1/]	4.6.3	3.5.4 (MIL-E-0022200E)
Covering, flaking & cracking[1/]	4.6.4	3.5.5 (MIL-E-0022200E)
Covering moisture[1/2/]	4.6.9	3.2.1 herein

1/ Required for qualification (QPL) only.

2/ Moisture content established at (QPL) inspection shall be reported in the certification of quality conformance inspection (See figure 5.)

4.4 Certificate of test. A certificate of quality conformance of the weld tests specified in 4.2.3, containing actual quantitative test and inspection results including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot included in a particular shipment, shall be furnished to the consignee with the shipment. The certification shall include test results from the quality conformance test. The minimum quality conformance test result data required shall be as listed in figure 5 and shall be submitted on the form as shown. In addition, the covering moisture content from QPL testing shall be included with the quality conformance test results (see figure 5). Covering moisture determination is not required for quality conformance inspection.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4)

5.1 Packaging (see 6.2).

5.1.1 Level A. All electrodes shall be packaged in class 1 unit containers in accordance with MIL-W-10430.

5.1.2 Level C. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and for time periods as specified in 3.5.8 in MIL-E-0022200E.

5.1.2.1 All electrode types shall be packaged in class 1 or 2 containers of MIL-W-10430. Unit container weight shall be in accordance with the manufacturer's commercial practice.

5.2 Packing (see 6.2).

5.2.1 Levels A and B. Electrodes shall be packed level A or B in accordance with MIL-W-10430, except that for levels A and B packing, palletization for other than cylindrical cans shall be in accordance with MIL-STD-147.

5.2.2 Level C. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

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5.3 Marking.

5.3.1 Interior packages. Shipment marking information shall be provided on interior packages in accordance with the manufacturer's commercial practice. In addition to the lot identification required by 4.4.6 of MIL-E-0022200E, the information shall include MIL type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation.

5.3.2 Exterior shipping containers and palletized unit loads. Shipment marking information shall be provided on exterior shipping containers and palletized unit loads in accordance with the manufacturer's commercial practice. In addition to the information required for interior packages as specified in 5.3.1, the information shall include shipping destination, stock number as listed by contract, and customer's order number, customer's item number, and customer's name.

6. NOTES

6.1 Intended use.

6.1.1 MIL-308-15 and 16. These electrodes are intended for the general welding of 18 percent chromium, 8 percent nickel, corrosion-resistant (stainless) steels.

6.1.2 MIL-308L15 and 16. These electrodes are intended for the general welding of 18 percent chromium, 8 percent nickel extra low carbon corrosion-resistant steel, used where intergranular corrosion must be limited but where corrosive conditions are not the most severe.

6.1.3 MIL-308HC-15 and 16. These electrodes are intended for applications where low magnetic permeability is of prime importance and where corrosion is not a factor.

6.1.4 MIL-309-15 and 16. These electrodes are intended for the joining of the 18-8 grades of corrosion-resistant steel to mild steel; special treatment steel, class B armor, and other low alloy air hardening steels, when specified; and carbon and straight chromium steels to austenitic grades. These electrodes are also for deposition of a corrosion-resistant overlay on carbon and low alloy steels.

6.1.5 MIL-309L-15 and 16. These electrodes are intended for similar applications as MIL-309-15 and 16 types, but where added resistance to intergranular corrosion is desired.

6.1.6 MIL-309CB-15 and 16. These electrodes are intended for similar applications as the MIL-309-15 and 16 types, but where expected corrosive conditions are more severe, or higher strength at elevated temperatures is desirable.

6.1.7 MIL-310-15 and 16. These electrodes are intended for the welding of 25 percent chromium, 20 percent nickel heat-resistant alloy steels used in high temperature service; special treatment steel, class B armor, and other low-alloy air hardening steels, where specified; and carbon and straight chromium steels to austenitic grades. These electrodes are also intended for deposition of a corrosion-resistant overlay on carbon and low alloy steels.

6.1.8 MIL-312-15 and 16. These electrodes are intended for the joining of the "super-alloys" to low-alloy (chromium-molybdenum, and chromium-nickel-molybdenum) steels.

6.1.9 MIL-316-15 and 16. These electrodes are intended for the joining of corrosion-resistant alloys of similar designations. The molybdenum content imparts increased resistance to corrosive effect of many organic acids, brine, sulfurous and sulfuric acid.

6.1.10 MIL-316L-15 and 16. These electrodes are intended for the welding of type 316L corrosion-resisting steel and under conditions similar to those under which type MIL-318 would be used.

6.1.11 MIL-317-15 and 16. These electrodes are intended for welding of alloys of similar composition for increased corrosion resistance to sulfuric and sulfurous acid and their salts.

6.1.12 MIL-318-15 and 16. These electrodes are intended for the welding of type 318 corrosion-resisting steel. Generally, uses are similar to those of type MIL-316 electrode, except that, where elevated temperatures and corrosive conditions favor intergranular corrosion and type 318 alloy is used, choice of stabilized type MIL-318 electrode is indicated.

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6.1.13 MIL-330-15 and 16. These electrodes are used in repair of alloy castings and welding of cast and wrought alloys of similar compositions which are generally used for heat-resisting properties of high temperatures.

6.1.14 MIL-347-15 and 16. These electrodes are intended for the general welding of 18-8 corrosion-resisting steels stabilized by columbium, which are used where conditions of welding or service make unstabilized base metals susceptible to carbide precipitation, with consequent intergranular corrosion. Use of these electrodes will not prevent intergranular corrosion alongside a weld in unstabilized 18-8 chromium-nickel base metals.

6.1.15 MIL-347HC-15 and 16. These electrodes are intended for applications where low magnetic permeability or heat resistance is of prime importance and where corrosion is not a factor.

6.1.16 MIL-349-15 and 16. These electrodes are intended for the joining of the "super-alloys" for applications requiring high strength at elevated temperatures.

6.1.17 MIL-16.8.2-15 and 16. These electrodes are intended for use in welding 316, 317, and 347 alloys for corrosion resisting and high temperature service in as-welded or stress-relieved condition.

6.2 Ordering data.

6.2.1 Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and size of electrode required (see 1.2).
(c) Standard stock numbers.
(d) That the right is reserved to restrict awards to bidders offering all classes and sizes in a single grade of electrodes purchased under this specification.
(e) For 3/32 or 1/8 size electrode, the length required (see table IV).
(f) Levels of packaging and packing required (see 5.1 and 5.2).

6.2.2 Electrodes should be ordered by the pound.

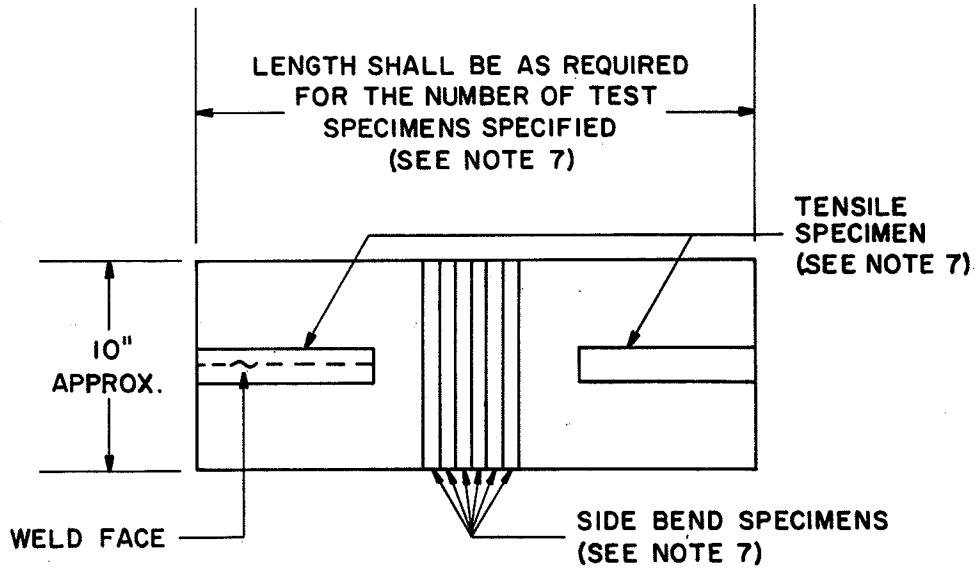
6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee. When receipt inspection testing types 308, 308L, 309, 309L, 309Cb and 347 for ferrite content, the consignee or user activity may consider ferrite test results from 2 to 12 percent as acceptable. Vendor tests shall meet the specified 4 to 9 percent ferrite content.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of QQ-S-766 do not apply when procured by the electrode manufacturer.

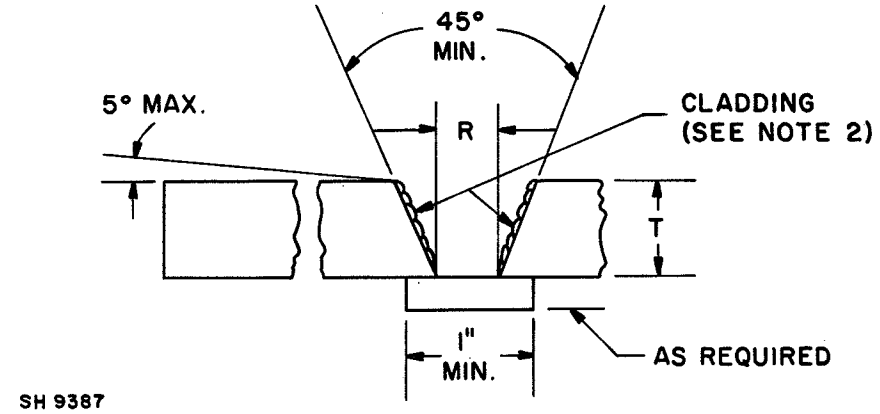
6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 3439-N171)

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LAYOUT OF TEST PLATE



GROOVE PREPARATION OF TEST PLATE

Electrode size (Inch)	Minimum thickness (T) (Inch)	Maximum Root opening (R) (Inch)	No. of layers 1/		Current (Minimum)	
			Minimum	Maximum	D.C. (Reverse-polarity) (Amperes)	A.C. (Amperes)
3/32	1/2	1/4	1/	1/	75	85
1/8	1/2	1/4	1/	1/	95	105
5/32	3/4	1/2	7	9	135	150
3/16	3/4	1/2	7	9	155	170
1/4	3/4	1/2	7	9	245	265
5/16	3/4	1/2	7	9	345	365

1/ Pass and layer sequence shall be reported.

Figure 1 - Welded joint for mechanical and bend tests (Test No. 3)

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Notes to figure 1:

1. The base material should match the chemistry of the electrode (same type designation in most instances). When base material matching the chemistry of the electrode is not available, AISI type 304 and 347 material may be substituted for the fabrication of tension and bend specimens provided that for tension specimens, cladding or buttering is accomplished. Base materials shall conform to QQ-S-766, ASTM A167 or ASTM A240.
2. If cladding is necessary, a minimum thickness of 1/8 inch shall be deposited on the plate edges comprising the groove. This may be done either prior to or after joint fit-up.
3. Preheat and interpass temperature shall not exceed 350°F.
4. Welding shall be accomplished in the flat position. Welds shall be deposited in layers using the 3 electrode core diameter maximum weave build up sequence. Record and report layer sequence as specified in figure 1. Layers should be approximately 1/8 inch thick and shall be deposited in such a manner that the final layer includes a reinforcement of standard proportions (1/32 inch to 1/8 inch). Welding currents shall be as shown in figure 1 and actual value shall be recorded and reported. Each layer shall be started at the finishing end of the preceding layer.
5. The weld reinforcement and backing strip shall be removed flush with base plate surfaces, after completion of welding. The assembly shall be radiographed in accordance with MIL-STD-271. Radiographs shall meet the acceptance standard of NAVSHIPS 0900-003-9000 as specified in 3.7.1 of MIL-E-0022200E.
6. Tensile specimens from 3/4 inch test plate shall be 0.505 inch diameter; those from 1/2 inch test plate shall be 0.250 inch diameter. For details of tensile specimens, see MIL-STD-418.
7. The number and type of mechanical tests shall be as follows:

Electrode type	Size (Inch)	Number of test specimens					
		Tensile[1]				Side Bend[1]	
		0.250 Inch		0.505 Inch			
		AW[2]	HT[2/3]	AW[2]	HT[2/3]	AW[2]	HT[2/3]
MIL-347, MIL-347HC, MIL-16.8.2	3/32 and 1/8	2	2	-	-	3	3
	5/32 to 5/16	-	-	2	2	3	3
All other	3/32 and 1/8	2	-	-	-	3	—
	5/32 to 5/16	-	-	2	-	3	—

[1] Tensile and side bend tests shall be conducted in accordance with MIL-STD-418.
[2] AW = As welded, HT = Heat treated.
[3] Heat treatment shall be as specified in note 8.

8. When heat treatment is required, the test specimens shall be heated to 1550°F., held for 1 hour and cooled. Heating and cooling rates shall not exceed 300°F. per hour.

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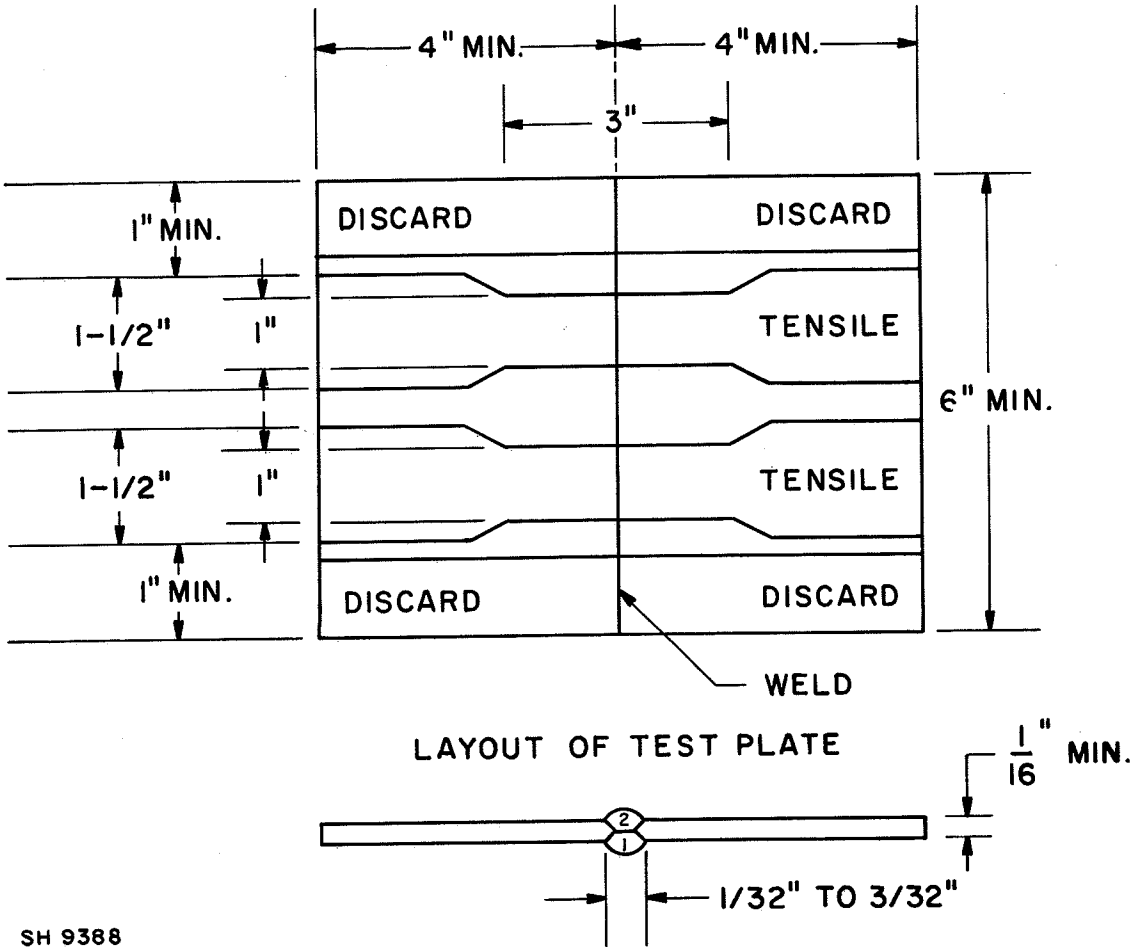
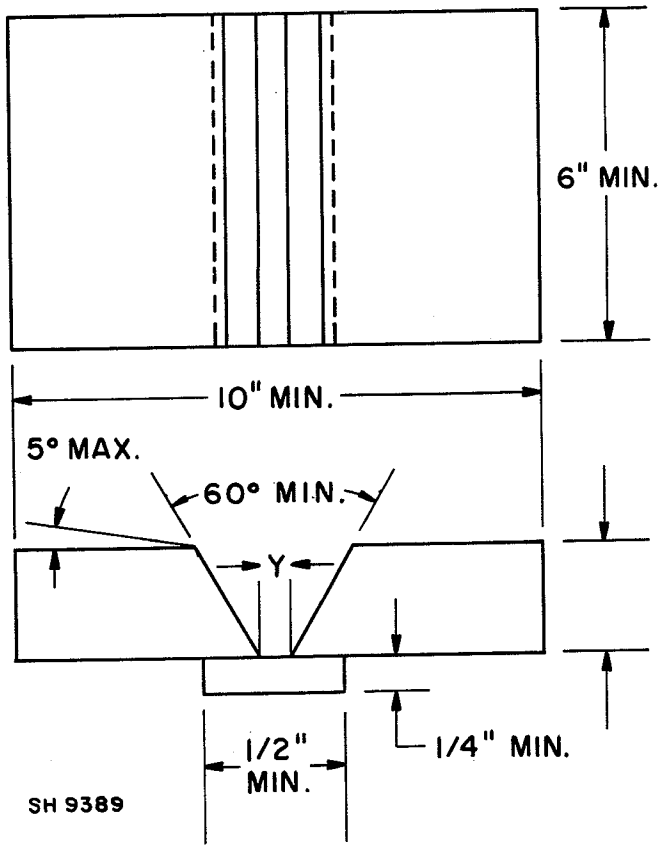


Figure 2 - Transverse weld tensile test for 1/16 inch and 5/64 inch electrode (Test No. 6)

Notes to figure 2:

1. Base material shall be as specified in note 1 to figure 1.
2. Welding shall be done in the flat position. Weld deposition shall be as shown. The backside of weld No. 1 shall be cleaned by grinding prior to depositing weld bead No. 2. Welding interpass temperature shall be 350°F. maximum.
3. The weld reinforcement shall be removed flush with the base material on both surfaces and the assembly shall be radiographed in accordance with MIL-STD-271. The radiograph shall meet the acceptance standards of NAVSHIPS 0900-003-9000 as specified in 3.7.1 of MIL-E-0022200E.
4. Two transverse weld tensile specimens shall be prepared as shown and tested in the as-welded condition in accordance with MIL-STD-418. Failure may be in weld or base material; however, to be acceptable the strength shall be equal to or greater than that specified in table II for the type electrode being tested.

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Electrode size	1/16 and 5/64 inch	3/32 inch	1/8 inch	5/32 inch
Thickness (T) Min.	1/8 inch	1/4 inch	3/8 inch	1/2 inch
Root opening (Y)*	3/16 inch	1/4 inch	5/16 inch	3/8 inch

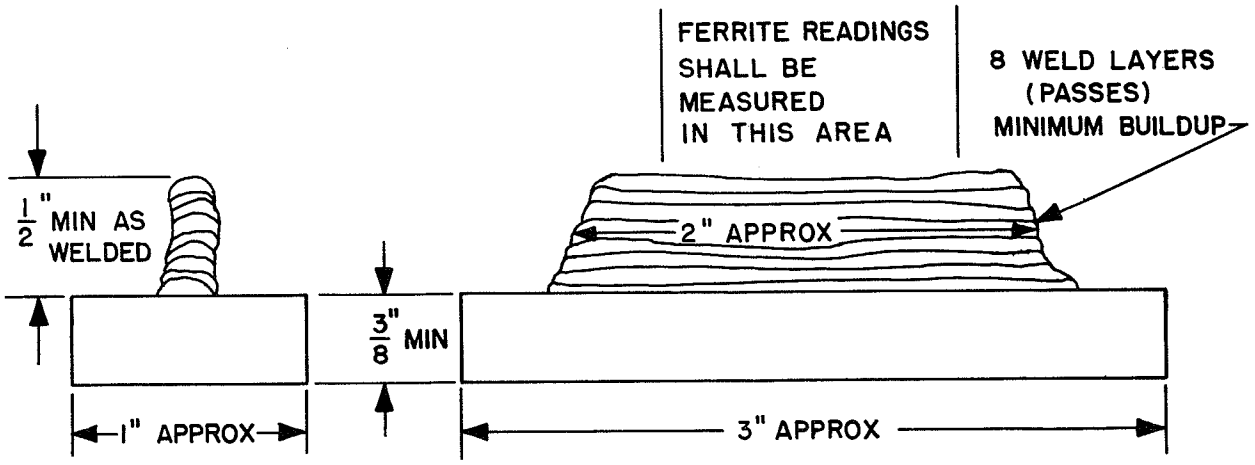
*Tolerance ± 1/16 inch

Figure 3 - Groove weld usability test (Test No. 8a)

Notes to figure 3:

1. Welding shall be accomplished in the vertical position.
2. Base metal shall be in accordance with any of the following AISI types: 304, 316, 316L and 347 conforming to QQ-S-766, ASTM A167 or ASTM A240.
3. The bevel edge preparation shall be machined or ground.
4. Welds shall be deposited in approximately 1/8 inch thick layers. Each weld bead shall contain a start in the area to be evaluated. Weld deposits shall be made using a 3 core wire diameter maximum weld weave.
5. The completed weld reinforcement shall be 3/16 inch maximum.
6. Preheat and interpass temperature shall be 60°F. minimum and 350°F. maximum.
7. After completion of the weld, the weld reinforcement and backing strip shall be removed flush with base plate surfaces and the assembly shall be X-rayed in accordance with MIL-STD-271.
8. The radiographs shall meet the acceptance standards of NAVSHIPS 0900-003-9000 as specified in 3.7.1 of MIL-E-0022200E.

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Electrode diameter (Inch)	Welding current (Amperes)	Arc voltage (Volts)	Electrode diameter (Inch)	Welding current (Amperes)	Arc voltage (Volts)
1/16	30-40	20-22	5/32	110-130	23-25
5/64	45-55	21-23	3/16	140-165	24-26
3/32	55-65	22-24	1/4	230-270	24-27
1/8	80-90	23-25	—	—	—

Figure 4 - Ferrite weld specimen (Weld Test No. 4)

Notes to figure 4:

1. The base material for the weld pad shall be Types 301, 302, or 304 conforming to QQ-S-766, ASTM A167 or ASTM A240.
2. The weld pad shall be built up by depositing single bead layers one on top of the other to a minimum height of 1/2 inch. The weld bead layers shall be deposited with a maximum weave width (electrode manipulation) of 2-1/2 times the core wire diameter, except the first several layers may be deposited with a wider weave.
3. The welding techniques shall comply with the above table of amperages and voltages. The weld stops and starts shall be located at the ends of the weld buildup.
4. The weld pad shall be cooled between passes by water quenching immediately after each pass, except the last two weld passes shall be air cooled below 800°F. prior to water quenching.
5. Each weld pass shall be cleaned prior to depositing the next weld bead.

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Notes to figure 4 (Cont'd):

6. After welding, the weld buildup shall be surface ground to 64 RMS finish to provide sufficient finished surface to make the required ferrite readings. The ground area need not be continuous along the specimen. The weld buildup shall be continuously water cooled during the grinding operation. The finished ground surface shall be approximately parallel with the plate surface and shall be located within the last two passes or layers of weld buildup. Machining and metallographic polishing sufficient to remove the cold working produced by machining may be used in lieu of surface grinding. The weld buildup should be water cooled during the machining operation.
7. A total of six ferrite readings shall be taken on the machined surface along the longitudinal axis of the weld bead. The readings should not be taken on either end where the weld craters (stop and start areas) are located. Reading should be taken in areas having dimensions at least 1-1/4 times the diameter of the magnet. Ferrite readings shall be taken by the magna-gage.
8. The six readings obtained shall be average to a single value for conversion to percent ferrite.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS[1/2/]

Manufacturer or Distributor _____

Address _____

Date _____

Specification MIL- _____

Type MIL- _____

Diameter & Length _____

Inspection Level _____

Lot. No. _____

Customer's Name _____

Customer's Order No. _____

Core Wire Heat No. _____

Lot Identification MIL-E-0022200E Para. _____

Chemical Analysis

Carbon _____

Chromium _____

Nickel _____

Molybdenum _____

Columbium _____

Tungsten _____

Chemistry was taken from: ☐ Chem Pad ☐ Groove Weld

X-Ray Results _____

Concentricity (%) _____

Covering Moisture (QPL) (%) _____

Ferrite Content

Percent ferrite (average) _____

Groove Weld Test

Test No.	8a	Chem Pad	3	6
Amperage	_____	_____	____	____

Grinding during 8a Test Plate Preparation[3/] Operator Error (Layer Nos.) _____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

[1/] Items not applicable to the type electrode involved or exempted by the procurement document shall be marked N/A.

[2/] This form is applicable to the quality conformance tests required for each lot and a copy shall be forwarded with each shipment.

[3/] See 3.5.7 and 3.9 of MIL-E-0022200E.

Figure 5

*U.S. GOVERNMENT PRINTING OFFICE: 1969-343-225/908

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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